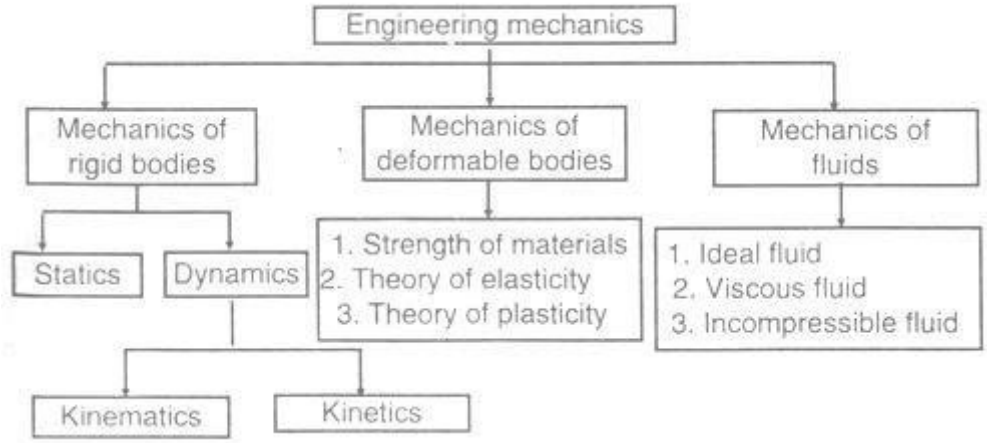
Subject: ENGINEERING MECHANICS

Unit 1: Introduction to Engineering Mechanics

Mechanics

It is defined as that branch of science, which describes and predicts the conditions of rest or motion of bodies under the action of forces. Engineering mechanics applies the principle of mechanics to design, taking into account the effects of forces.



Statics- Branch of EM, which deals with the forces and their effects, while acting upon the bodies at rest.

Dynamics- Branch EM, which deals with the forces of their effects, while acting upon the bodies in motion.

Kinematics - Branch of Dynamics, which deals with the bodies in motion, without any reference to forces which are responsible for the motion.

Kinetics - Branch of dynamics which deals with the bodies in motion due to the application of forces.

Some Important Definitions

1. Space

In Engineering Mechanics, *space* refers to the region in which a body exists and moves. It is the three-dimensional framework that defines the position of a particle or body using coordinates (x, y, z).

- Space allows engineers to describe location, distance, and direction.
- All physical events such as motion, force application, and deformation occur within space.

- In mechanics, space is assumed to be continuous and measurable. Understanding space is essential for representing forces, displacement, velocity, and acceleration in vector form.

2. Time

Time is a fundamental quantity that measures the sequence and duration of events.

- It is used to describe motion, velocity, acceleration, and dynamic behavior of bodies.
- In statics, time does not influence the system (forces are constant).
- In dynamics, time is crucial because motion and forces vary with time.
- SI unit of time is the **second (s)**.

Time helps in analyzing how a system changes, especially in motion-related problems.

3. Mass

Mass is the quantity of matter contained in a body. It is a measure of a body's inertia—its resistance to change in motion.

- SI unit: **kilogram (kg)**
- Mass is independent of location and remains constant everywhere.
- It differs from weight:

$$\text{Weight} = \text{Mass} * g$$

- In mechanics, mass plays a vital role in dynamics, especially in Newton's laws of motion.

4. Particle

A *particle* is an idealized body having mass but negligible size.

- Used when the dimensions of a body are very small compared to the distances involved.
- Shape and size are ignored, but mass is retained.
- Examples:
  - A vehicle traveling between cities
  - A projectile in motion

In mechanics, many problems are simplified by treating bodies as particles when rotation and deformation are not significant.

5. Flexible Body

A *flexible body* is one that undergoes large deformations when subjected to forces.

- The shape and size change significantly under load.
- Examples:
  - Ropes
  - Cables
  - Chains
  - Belts

Characteristics:

- Cannot resist bending moments.
- Can only carry tensile forces.
- Used in problems involving pulleys, hoists, and lifting machines.

Flexible bodies are important in applied mechanics where tension members transmit forces between rigid bodies.

6. Rigid body

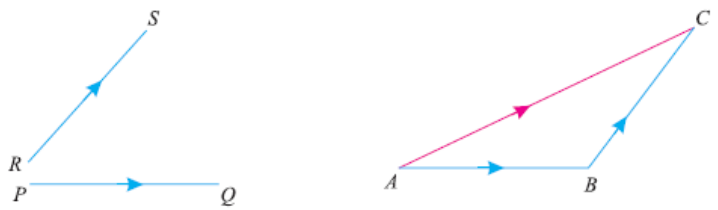
A rigid body is defined as a definite quantity of matter, the parts of which are fixed in position relative to each other. Physical bodies are never absolutely but deform slightly under the action of loads. If the deformation is negligible as compared to its size, the body is termed as rigid.

SCALAR QUANTITIES

The scalar quantities (or sometimes known as scalars) are those quantities which have magnitude only such as length, mass, time, distance, volume, density, temperature, speed etc.

VECTOR QUANTITIES

The vector quantities (or sometimes known as vectors) are those quantities which have both magnitude and direction such as force, displacement, velocity, acceleration, momentum etc. Following are the importantfeatures of vector quantities



(*a*) Vectors *PQ* and *RS* (*b*) Addition of vectors

SYSTEM OF UNITS

1. Fundamental Units

These are **basic units** that are **independent** and cannot be expressed in terms of other units.

In mechanics, there are **three fundamental quantities**:

Quantity	SI Unit	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s

2. Derived Units

These are units **derived from fundamental units** using mathematical relations.

Common Derived Units in Mechanics (SI)

Quantity	Formula	Unit	Symbol
Area	Length × Length	m^2	
Volume	$Length^3$	m^3	

1. CGS System (Centimetre–Gram–Second)

Basic units:

- Length → centimetre (cm)
- Mass → gram (g)
- Time → second (s)

Derived units:

- Force → dyne
- Work/Energy → erg
- Pressure → barye

Use: Mostly in theoretical physics and older literature.

2. MKS System (Metre–Kilogram–Second)

Basic units:

- Length → metre (m)
- Mass → kilogram (kg)

- Time → second (s)

Derived units:

- Force → newton (N)
- Work/Energy → joule (J)
- Pressure → pascal (Pa)

Use: Foundation of the modern SI system.

3. FPS System (Foot–Pound–Second)

Basic units:

- Length → foot (ft)
- Mass → pound (lb)
- Time → second (s)

Derived units:

- Force → pound-force (lbf)
- Work → foot-pound (ft·lb)

Use: Engineering practices in the USA.

4. SI System (International System of Units)

This is the **most widely used system today**.

Basic units in mechanics:

- Length → metre (m)
- Mass → kilogram (kg)
- Time → second (s)

Property	Unit	Abbreviation
Length	meter	m
Mass	kilogram	kg
Time	seconds	s
Amount	mole	mol
Temperature	kelvin	K
electric current	ampere	amp
luminous intensity	candella	cd

Unit 1: Introduction to Engineering Mechanics

Force

A **force** is a **push or a pull** that can change the **shape, size, direction, speed, or state of rest/motion** of an object.
If something **moves, stops, changes direction, or deforms**, a force is involved.

Effects of Force

A force can:

1. **Move a stationary object**
   Example: Pushing a stationary trolley.
2. **Stop a moving object**
   Example: Applying brakes on a bicycle.
3. **Change speed** (increase or decrease)
   Example: Kicking a football harder makes it go faster.
4. **Change direction of motion**
   Example: Bat hitting a cricket ball.
5. **Change shape or size**
   Example: Stretching a rubber band.

CHARACTERISTICS OF A FORCE

In order to determine the effects of a force, acting on a body, we must know the following

Characteristics of a force:

1. Magnitude of the force (i.e., 100 N, 50 N, 20 kN, 5 kN, etc.)

2. The direction of the line, along which the force acts (i.e., along OX, OY, at 30° North of East etc.). It is also known as line of action of the force.

3. Nature of the force (i.e., whether the force is push or pull). This is denoted by placing an arrow head on the line of action of the force.

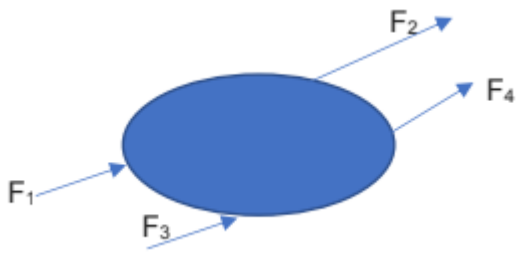
4. The point at which (or through which) the force acts on the body.

PRINCIPLE OF PHYSICAL INDEPENDENCE OF FORCES

The Principle of Physical Independence of Forces states that when multiple forces act simultaneously on a body, each force produces its own specific acceleration independent of the presence of other forces. The net acceleration is the vector sum of these individual accelerations, meaning one force does not affect the magnitude or direction of another.

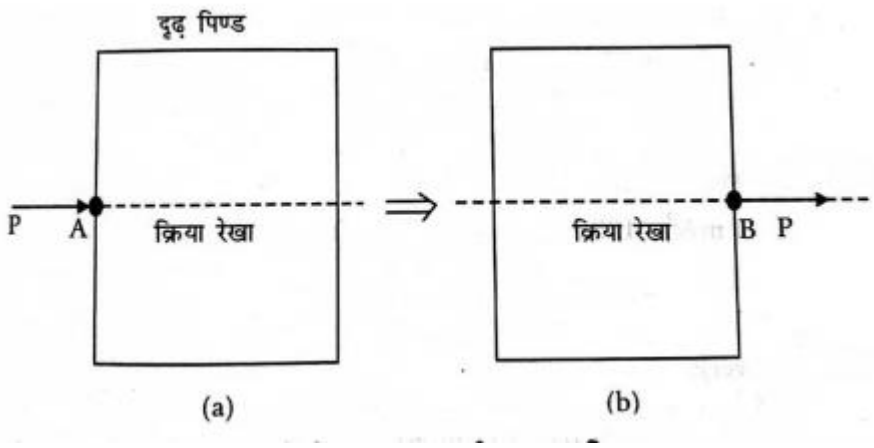
Key Aspects of the Principle

- **Independent Action:** Each force acts as if it were the only force present.
- **Vector Superposition:** The combined effect (resultant force) is the vector sum of all individual forces acting independently.
- **Consequence of Newton's First Law:** This principle is a fundamental consequence of Newton's First Law of Motion, which implies that forces act independently without influencing each other's existence.
- **Applications:** It is essential for resolving problems in mechanics, such as analyzing projectile motion by separating horizontal and vertical components.



PRINCIPLE OF TRANSMISSIBILITY OF FORCES

It states, "If a force acts at any point on a †rigid body, it may also be considered to act at any other point on its line of action, provided this point is rigidly connected with the body."



Multiples and Submultiples of SI units

Prefix	Symbol	Multiplying Factor	
exa	E	10^{18}	1 000 000 000 000 000 000
peta	P	10^{15}	1 000 000 000 000 000
tera	T	10^{12}	1 000 000 000 000
giga	G	10^{9}	1 000 000 000
mega	M	10^{6}	1 000 000
kilo	k	10^{3}	1 000
hecto*	h	10^{2}	100
deca*	da	10	10
deci*	d	10^{-1}	0.1
centi	c	10^{-2}	0.01
milli	m	10^{-3}	0.001
micro	u	10^{-6}	0.000 001
nano	n	10^{-9}	0.000 000 001
pico	p	10^{-12}	0.000 000 000 001
femto	f	10^{-15}	0.000 000 000 000 001
atto	a	10^{-18}	0.000 000 000 000 000 001